PROGRESS REPORT

Name: Nathan Walker

Project Title: IFEC Audio Amplifier

Tell us about last week's goals. How did they go?

To get all of the possible points for this section, write at least three sentences for each goal. Describe what each goal was and how you accomplished it (or what happened to prevent you from accomplishing it).

Goal #1: In the text box below, describe the goal and the work you did to accomplish it.

We had to meet as a team and discuss what we needed to do to fix the boards. We talked about some of the hardware problems that we could fix to improve filter design. We also discussed some of the software that we could change to improve controls.

Status: ☐ Complete ☒ Partially Complete ☐ No Progress

Goal #2: In the text box below, describe the goal and the work you did to accomplish it.

Change the Inductor air gap to change the inductance in the filter. This requires us to change the shim or the material in between the cores to a smaller airgap. However, due to the smaller air gap we need something to secure the cores together to the board.

Status: ☐ Complete ☒ Partially Complete ☐ No Progress

Goal #3: In the text box below, describe the goal and the work you did to accomplish it.

Seeing what is damaged on the boards and fix or replace them. It is still a little too early to determine what is broken as we want the inductor figured out first. After we fix the inductor we will test and make sure the switches are operating correctly. If it seems to not be working properly we will desolder the switches off and replace them.

Status: ☐ Complete ☒ Partially Complete ☒ No Progress

Set three goals to work on next week.

Progress reports are **individual** assignments. That means you will be setting goals that **you yourself** intend to accomplish this week. Be specific; set goals that can be completed in one or two weeks.

As you work on your goals, make sure to take at least one picture or screenshot to show what you have done. The picture or screenshot will need to be submitted in your next progress report.

To get all of the possible points for this section, write at least three sentences for each goal. Describe what the goal is and how you plan to accomplish it.

Goal #1:

Finish making changes in the inductor design. This requires us to use glue or something to secure the inductor cores so they are not wobbling around. After this is done we will be using an LCR meter to measure the inductance to make sure it is within the correct ranges.

Goal #2:

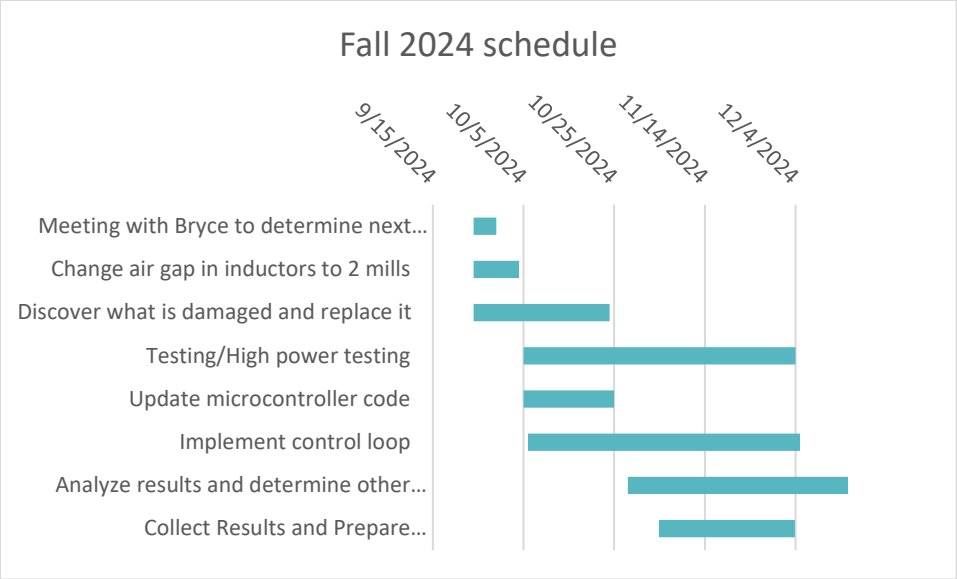
Seeing what Damaged on the boards and what components needs to be replaced. Some components that we would need to check is the switches, the sensing circuits, and the microcontroller. These circuits are the most sensitive on the board to damage so using oscilloscopes and probing pins to make sure they are outputting what we are expecting.

Goal #3:

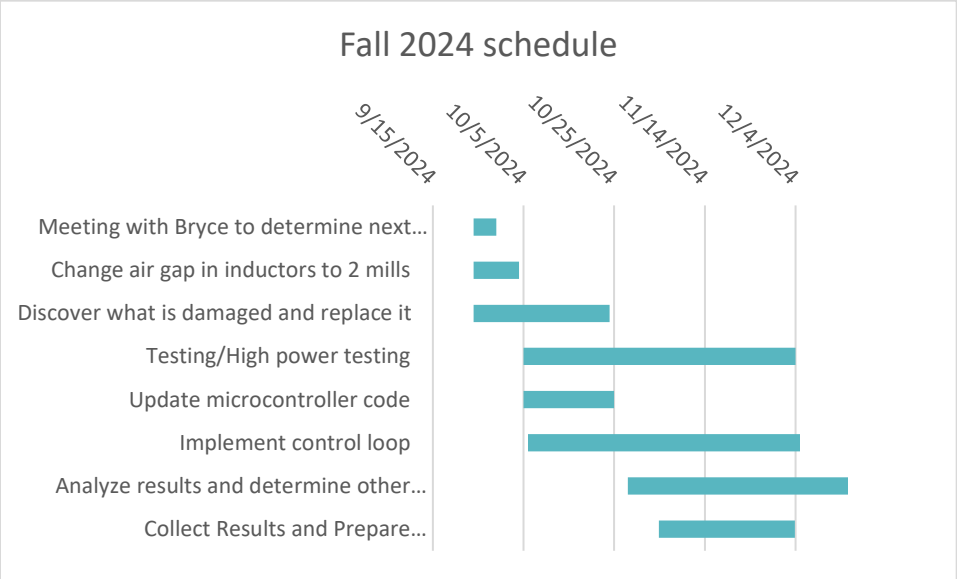
Testing the boards and overall circuits. Once we change the inductor and replace any damaged parts we will want to perform tests. The tests will start with first testing the boards to ensure functionality and then low power testing. Once we confirm those two tests we will move onto increasing the power and testing it.

Visual Communication

Each report must be accompanied by an updated schedule in the form of a Gantt chart. You need to include the version from the previous progress report **and** the updated chart for next week. Paste them below or save the charts as pdfs and upload them separately.



First Gantt chart



Updated Gantt chart

Each report must also be accompanied by evidence of your work. This can be a photo, screenshot, graph, chart, code snippet, data table, or any other visual evidence that we discuss in class. Paste it below or upload it as a separate file.

The picture shows work being done to measure the inductor on the board.

